Paul Parisot

Software Engineer - Embedded Linux & C++ Systems

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About

Systems software engineer experienced in low-level Linux environments, hardware communication boundaries, and high-performance applications.
Strong understanding of runtime memory layouts, custom allocators, concurrent programming, and containerized architectures.
Pragmatic engineer proficient in system profiling, automated debugging, and translating hardware constraints into robust software solutions.

Skills

Core Systems: C, C++ (C++14/17/20), Go, C# (Unity), x86/ARM Assembly stubs
OS & Memory: Memory Management (`heap, stack, shm`), POSIX threads, Inter-Process Communication (IPC)
DevTools & Infrastructure: Linux, CMake, GCC, GDB, Valgrind, Docker Engine, Git, Cross-Compilation
Automation & Scripting: Python scripting, Bash, System Automation, Code Auditing

Professional Experience

Faurecia / Forvia - Software Engineer (Systems & Tooling)

- Developed and integrated low-level C++ software modules dedicated to hardware driver communication boundaries for embedded applications.
- Engineered embedded C# applications using the Unity engine for real-time sub-system visualization and testing tools.
- Managed strict execution and resource constraints on targeted physical architectures.
- Worked within an industrial Agile/Scrum environment, tracking functional tasks and technical workflows via **Jira** and **Git**.

Gno.land - Core Software Engineer (Distributed Runtime Engines)

- Refactored core runtime sub-systems, focusing on multi-threaded thread safety and resource-efficient concurrency.
- Isolated and resolved **6 critical execution bugs** (race conditions, memory leaks), committing over 10,000 lines of production code.
- Designed full integration test suites to validate complex system-level protocols and distributed communications.

Epitech / IONIS - Systems Programming Instructor & Code Auditor

- Instructed **400+ students** in advanced C/C++ memory management, low-level pointers, and Unix system calls.
- Conducted exhaustive code audits to eliminate undefined behaviors, memory corruption, and unhandled system signals.

Systems & Low-Level Technical Projects

rodox – Low-Latency Linux Systems Engine

Engineered a modular runtime architecture under Linux utilizing a data-oriented layout (ECS) to optimize CPU cache lines.
Developed custom **static pool allocators** to entirely bypass OS-level heap allocation latencies (`malloc/new`).
Implemented high-performance asynchronous networking and communication layers using `Boost.Asio` over TCP. Profiled and hardened code integrity using **GDB** and **Valgrind** to guarantee absolute deterministic execution.

Skills: C++, Boost.Asio, Memory Management, Linux System APIs, Profiling (Valgrind), CMake, Concurrency

Sledge – Containerized Orchestration & Core Backend

Architected and developed the complete backend infrastructure in Go optimized for high-volume data operations.
Implemented runtime container management logic by leveraging the Docker API to spawn and monitor environment tasks dynamically.
Authored custom automation scripts in **Python** to streamline pipeline testing, execution tracking, and stability validation.

Skills: Go, Docker API, Python Scripting, Data Streaming, Linux Environments, System Automation

cclass – Runtime Memory Mutation Engine

Designed a pure C framework that dynamically modifies compiled assembly instructions inside raw memory pages.
Utilized `mmap` system calls with precise protection flags to bypass traditional virtual table dispatch overheads.
Enforced strict memory alignment rules to maintain absolute stability across underlying hardware execution targets.

Skills: C, Linux Systems, x86 Assembly, Memory Safety, Low-level Debugging

Education

Epitech, Paris 2022 – 2027
Master's Degree in Information Technology (Master's level diploma)
Level-7 Certification - Expert in Software Engineering
Specialization: Systems Architecture

McGill University, Montréal 2025 – 2026
Certificate in Management
Technical Leadership & Strategy